

Uptake of EDTA-complexed Pb, Cd and Fe by solution- and sand-cultured *Brassica juncea*

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Abstract Direct plant uptake of metals bound to chelating agents has important implications for metal uptake and the free-ion activity model. Uptake of hydrophilic solutes such as metal–EDTA complexes is believed to occur via bypass apoplastic flow, but many questions remain about the relative importance and selectivity of this pathway. In this study, *Brassica juncea* (Indian mustard) plants grown in solution- and sand-culture conditions were exposed to metal–EDTA complexes and to PTS, a hydrophilic fluorescent dye previously used as a tracer of apoplastic flow. The results suggest that there are two general phases of solute uptake. Under normal conditions, xylem sap solute concentrations are relatively low (i.e., <0.5% of concentration in solution) and there is a high degree of selectivity among different solutes, while under

conditions of stress, xylem sap concentrations are significantly higher (i.e., >3% of concentration in solution) and the selectivity among solutes is less. In healthy plants, xylem sap metal–EDTA concentrations were generally an order of magnitude higher than those of PTS and differences among complexes were observed, with CdEDTA²⁻ exhibiting slightly higher xylem sap concentrations than PbEDTA²⁻ or FeEDTA⁻. Metal–EDTA complexes were found to dominate xylem sap metal speciation and the fraction of metal in xylem sap present as metal–EDTA was greater for non-nutrient metals (Pb, Cd) than for the nutrient metal Fe. Despite differences in root morphology between plants grown under solution- and sand-culture conditions, uptake of solutes was similar under both sets of growth conditions.

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Abbreviations

EDTA	ethylenediaminetetraacetic acid
PTS	8-hydroxy-1,3,6-pyrenetrisulfonic acid
EDDHA	ethylenediaminedi(<i>o</i> -hydroxyphenylacetic) acid
MES	2-(4-morpholino)ethanesulfonic acid
HBED	N,N'-di-(2-hydroxybenzoyl)-ethylenediamine-N,N'-diacetic acid

Introduction

The accumulation of toxic metals such as lead and cadmium by plants is a potential hazard for human and ecosystem health. However, it is difficult to predict metal accumulation by plants because the mechanisms that control this process are not fully understood. Predictions of metal uptake often rely on the free-ion activity model (FIAM) (Campbell 1995), which states that metal uptake and toxicity are most closely related to the free-ion activity of metal in solution. Indeed, a number of hydroponic and soil experiments have demonstrated a strong correlation between free-ion activity and plant uptake of metals (Bingham et al. 1983; Cabrera et al. 1988; Checkai et al. 1987; Pavan and Bingham 1982; Sauvé et al. 1996). However, a number of exceptions to the FIAM have been observed (Bell et al. 1991; McLaughlin et al., 1997, 1998; Parker et al. 2001; Smolders et al. 1998; Smolders and McLaughlin 1996). For instance, in some hydroponic experiments, metal uptake has been found to increase with increasing total metal concentrations even when constant free-ion activities were maintained (e.g., Bell et al. 1991; McLaughlin et al. 1997).

One major shortcoming of the FIAM is that it fails to account for the fact that intact metal–ligand complexes can be absorbed into plant roots and translocated to shoots. When chelating agents were first introduced into plant nutrition to overcome iron deficiency, Fe complexes with ^{14}C - and ^{15}N -labelled metal-binding ligands such as EDTA (Beckett and Anderson 1972; Wallace et al. 1955) and EDDHA (Hill-Cottingham and Lloyd-Jones 1965; Römhelt and Marschner 1981) were observed in the shoots of plants. Crowdy and Tanton (1970) reported the presence of PbEDTA^{2-} in the roots and shoots of wheat seedlings. Recent interest in using EDTA and other chelating agents to enhance phytoextraction of toxic metals has prompted further inquiry into the role of chelating agents in metal uptake. Addition of EDTA to metal-contaminated soils has been shown to enhance the transport of Pb from roots to shoots by more than two orders of magnitude compared to plants grown in the same soil without chelate addition (Grčman et al. 2001). The development of more sophisticated

analytical techniques has led to further evidence for the translocation of intact metal–chelate complexes within plants. Uptake of metal–EDTA complexes has been observed in a variety of plant species including *Brassica juncea* (Vassil et al. 1998), *Phaseolus vulgaris* (Sarret et al. 2001), *Hordeum vulgare* (Collins et al. 2001) and *Solanum tuberosum* (Collins et al. 2002). While these studies demonstrate that EDTA-bound metals can be transported to the shoots of plants, important questions remain about the mechanisms of uptake and the relative contribution of chelated metal uptake to overall metal uptake.

Transport of metal–EDTA complexes into plant roots is often thought to occur through the root apoplasm since large hydrophilic compounds, such as PbEDTA^{2-} , are unlikely to cross cell membranes (Bell et al. 1991; Crowdy and Tanton 1970; Tanton and Crowdy 1971). Thus, the movement of such solutes into the xylem is generally blocked by the endodermal Casparian band. However, the Casparian band has been found to be an imperfect barrier to solute transport, most notably because of incomplete development at root tips and breaks at points of lateral root growth, providing a “bypass flow” pathway for solutes in the apoplasm to be transported to the xylem (Yeo et al. 1987). Ranathunge et al. (2005) observed $\text{Fe}(\text{CN})_6^{4-}$ and Cu^{2+} movement across the Casparian band of corn and rice roots, most notably near the locations of lateral root emergence. Fluorescent tracers also have been used to visualize water and solute transport in roots. For instance, PTS (8-hydroxy-1,3,6-pyrenetrisulfonic acid) has been used as a tracer of apoplastic flow in red pine seedlings (Hanson et al. 1985), mangroves (Moon et al. 1986), rice (Yeo et al. 1987), cotton (Skinner and Radin 1994) and maize (Zimmermann and Steudle 1998). While several authors have rejected its use as a tracer of apoplastic water flow because of the substantial differences in diameter and membrane permeability between PTS and water (e.g., Steudle and Peterson 1998; Zimmermann and Steudle 1998), xylem sap PTS concentrations were found to be proportional to Na^+ concentrations in rice plant xylem sap (Yadav et al. 1996). Consequently, PTS also may be a reasonable tracer for larger hydrophilic solutes, including metal–

EDTA complexes. Although apoplastic bypass flow has long been recognized as the likely transport pathway, few studies have quantitatively addressed the relative importance of this pathway as a route of metal uptake and the selectivity of this pathway for various solutes and effects of growth conditions and stress.

This research was designed to assess the contribution of metal–EDTA uptake to overall metal uptake and to understand the mechanisms and pathways involved. Metal–EDTA concentrations were measured directly in the xylem sap of *B. juncea* plants exposed to various combinations of metal–EDTA complexes to quantify the uptake of intact metal–EDTA complexes by roots. In addition, uptake of metal–EDTA complexes was compared to that of PTS to determine if the two compounds demonstrated similar patterns of uptake. The results provide quantitative information about the relative ability of metal–EDTA complexes to be taken up through plant roots and evidence for interpreting the mechanisms involved. The effects of mechanical impedance and root morphology also were explored in a side-by-side comparison of metal–EDTA uptake using solution- and sand-culture methods.

Materials and methods

Growth and exposure conditions

Solution-grown plants exposed to one or more metal–EDTA complexes

Brassica juncea seeds (various cultivars, Northern Central Regional Plant Introduction Station, Ames, IA) were surface sterilized with 3% NaOCl for 30 min and germinated on 1.2% agar growth medium (modified from Vassil et al. 1998). Seeds were germinated in the dark for 24 hours before transfer to a light chamber (containing full-spectrum fluorescent lights) with a photosynthetic photon flux of 50–130 $\mu\text{mol m}^{-2} \text{s}^{-1}$ from full-spectrum fluorescent grow lights on a 16/8 hour light/dark photoperiod at 22–30°C. Humidity was not controlled. After approximately one week, healthy seedlings were trans-

ferred to 3.6-L polyethylene buckets (1 plant per bucket) with 3 l of constantly aerated chelator-buffered nutrient solution (pH 6, 2 mM MES, 2 mM KNO_3 , 2 mM $\text{Ca}(\text{NO}_3)_2$, 0.5 mM MgSO_4 , 0.2 mM NaH_2PO_4 , 10 μM H_3BO_3 , 0.1 μM Na_2MoO_4 , 2 M ZnCl_2 , 1 M CuCl_2 , 0.5 M MnCl_2 , 10 μM $\text{Fe}(\text{NO}_3)_3$, 55 M K_2EDTA). Black buckets and lids were used to minimize algal growth. The nutrient solutions were changed weekly or more frequently as necessary.

After an additional 6–8 weeks of growth (aboveground biomass generally 4–8 g D.W.), plants were transferred to buckets containing 3 l of experimental solutions, which consisted of the nutrients listed above as well as 50–1200 μM metal–EDTA. Concentrated solutions of EDTA in a 1:1 molar ratio with Cd, Pb and/or Fe were prepared separately and allowed to equilibrate overnight before addition to the nutrient solution. All experimental solutions contained 45–100 μM molar excess EDTA (above the total concentration of trace metals) to buffer micronutrient free ion activities. In solutions where uptake of FeEDTA^- was not being tested, 10 μM HBED (N,N'-di-(2-hydroxybenzoyl)-ethylenediamine-N,N'-diacetic acid) was added to buffer $[\text{Fe}^{3+}]$. HBED is highly selective with respect to its ability to form complexes with Fe and should not alter the speciation of other metals (Chaney 1988). A 1 M KOH solution was used to adjust the pH to 6.1 ± 0.2 . To measure apoplastic bypass flow, 380 μM (0.2 g/l) PTS (8-hydroxy-1,3,6-pyrenetrisulfonic acid), a hydrophilic fluorescent dye, was added to some experimental solutions. PTS has previously been used as a tracer of apoplastic flow (e.g., Hanson et al. 1985). Preliminary experiments showed that this concentration of PTS did not cause deleterious effects on plants (data not shown).

After 24–60 h of exposure to the experimental solutions, the plants were decapitated by slicing the stem just below the lowest true leaves. Xylem sap was collected for 2–3 h at atmospheric pressure and stored at 5°C until analysis. Samples of experimental solutions were collected before and after plant growth to measure changes in metal and EDTA concentrations over the course of the experiments. Metal and EDTA concentrations generally increased 10–20% during plant

exposure due to transpiration of water from the solutions. Following harvest, leaf samples were weighed, dried at 100°C for 24 h, and reweighed to determine water content. Bulk transpiration was estimated by measuring the volume of solution lost over the course of the experiment. Evaporation from buckets without plants was negligible. Transpiration rates were normalized to total leaf dry weight.

In some plants, visible damage occurred to the leaves following exposure to the metal–EDTA solutions. This damage was characterized by light brown or translucent necrotic spots on the leaves and limpness. The degree of damage was characterized by visual inspection as “minimal” (no visible changes from pre-exposure), “moderate” (leaves had some necrotic spots, covering less than approximately 5% of total leaf area), and “severe” (significant damage to leaves, necrotic spots covering more than 5% of leaf area).

Solution- and sand-grown plants exposed to metal–EDTA

Experiments comparing solute uptake in solution- and sand-cultured plants were performed in a greenhouse at the University of California, Riverside in August 2002. *Brassica juncea* plants (var. Southern Giant Curled) were grown simultaneously in the two culture systems. Solution-cultured plants were first germinated on filter paper prior to transfer to black polyethylene buckets containing 8.25 l of constantly aerated nutrient solution (as described above). For sand culture, *B. juncea* seeds were sown directly in buckets containing 11.4 kg silica sand (grit size 16, Oglebay Norton, CA). Each pair of sand-filled buckets received nutrient solution from one 120-l reservoir circulated through for five minutes, five times per day. All buckets were thinned to contain four plants each. After three weeks of additional growth, plants were exposed to nutrient-containing experimental solutions, containing either PbEDTA²⁻ alone (100 or 1000 μM) or equimolar PbEDTA²⁻ and FeEDTA⁻ (50 or 500 μM each). Equilibrium calculations using GEOCHEM indicated that more than 99.99% of Pb and Fe was bound to EDTA. The metal and EDTA were

allowed to equilibrate overnight prior to addition of other nutrients and commencement of exposure. All solutions contained 45 μM excess K₂EDTA but did not include HBED. For each treatment, there were two sand-filled buckets, fed by a common 120-l reservoir, and two solution-filled buckets. Plants were exposed to metal–EDTA solutions for 2 days. Mean high and low temperatures in the greenhouse were 32 and 22°C and midday photosynthetically active radiation (PAR) averaged 900 $\mu\text{mol m}^{-2} \text{s}^{-1}$.

Before harvesting, the stomatal conductance and transpiration rates of a subset of the plants were measured using a LiCor LI-1600 Steady State Porometer (Lincoln, NE). At approximately 11:00 am, measurements were taken on newest fully-extended leaves from control plants and plants in the 500 μM FeEDTA⁻ and 500 μM PbEDTA²⁻ exposure for both sand and solution conditions. Following harvest, xylem sap was collected as described previously. Above-ground biomass was dried at 60°C for 48 h prior to analysis. Aliquots of nutrient solution were collected before and after plant exposure and subsequently analyzed for changes in metal–EDTA concentrations.

Chemicals

All reagents were prepared using deionized water from a Nanopure II apparatus (Barnstead). MES (2-(4-morpholino)ethanesulfonic acid), K₂EDTA (ethylenediaminetetraacetic acid, dipotassium salt) and cetyltrimethyl-ammonium bromide were obtained from Sigma-Aldrich. HBED was obtained from Strem Chemicals. PTS was obtained from Spectrum Quality Products. All other chemicals and all HPLC-grade solvents were obtained from Fisher Scientific.

Analytical techniques

Dried shoot biomass was ground and subsamples were digested with concentrated HNO₃ and 30% H₂O₂ (Vassil et al. 1998). Total metal concentrations were determined using graphite furnace atomic absorption spectrometry (Perkin Elmer 3300) after diluting the extracts as necessary to achieve concentrations in the range of the

calibration curves. Xylem sap samples were diluted to a final concentration of 0.1 N HNO₃, while shoot digests often contained higher HNO₃ concentrations. All samples were analyzed in duplicate. The accuracy of the measurements was verified with a standard reference solution of trace elements (NIST SRM 1640d, Trace Elements in Water) as well as with methodological blanks and spike recoveries. The average errors from the reported NIST SRM concentrations were 10% (Pb), 6% (Cd), and 15% (Fe). Metal concentrations in digestion blanks were generally below the detection limit or negligible compared to sample concentrations. Metal concentrations in digestion spike recoveries ranged from 89 to 106% of the expected concentration.

Total EDTA and FeEDTA⁻ in xylem sap samples were determined using ion-pair high pressure liquid chromatography (HPLC) and UV detection (Nowack et al. 1996). Total EDTA was determined by pre-analysis conversion of the metal-EDTA complexes into FeEDTA⁻ by adding 10% (v/v) concentrated iron solution (40 mM Fe(NO₃)₃, 200 mM formate buffer, pH 3.3) to samples. Samples were analyzed for FeEDTA⁻ with a 25-cm RP LC-18 column (Supelco) using a mobile phase consisting of 9% acetonitrile, 20 mM formate buffer, 1 mM tetrabutylammonium bromide followed by UV detection at 258 nm. The detection limit of EDTA in xylem sap samples was approximately 0.1 μM. Identity of peaks was confirmed by ensuring that the ratio of absorbance at 258 nm to absorbance at 300 nm agreed with those of standards.

PbEDTA²⁻ and CdEDTA²⁻ were determined using an ion-exchange HPLC method (Bedsworth and Sedlak 2001). Metal-EDTA complexes were separated with an aqueous eluent (6 mM Na₂SO₄, 1 mM HEPES, pH 7) on a 25-cm RP LC-18 column (Supelco) that had been coated with the surfactant cetyltrimethylammonium bromide, followed by post-column conversion to FeEDTA⁻ and detection at 258 nm. For this method, the detection limit for surface water and wastewater effluent samples is approximately 0.07 μM. For xylem sap samples, the detection limit ranged from 0.2 to 0.5 μM. The identity of peaks was confirmed by checking that the ratio of absorbance at 258 nm to absorbance at 300 nm agreed

with those of standards. In some cases, quantification was not possible due to the presence of coeluting peaks.

PTS concentrations were determined by integrating peak areas recorded after passing 50 μl samples through a fluorescence detector (Shimadzu RF-10AXL, excitation λ = 403 nm, emission λ = 513 nm). The samples were passed through the detector without an HPLC column using the same eluent as in the EDTA speciation method at 1.5 ml/min. Since the fluorescence of PTS changes as a function of pH, using a buffered eluent was important to ensure that all samples and standards were analyzed at the same pH. The fluorescence of xylem sap from a control plant (not exposed to PTS) was subtracted from the fluorescence of samples to account for the inherent fluorescence of the xylem sap. PTS concentrations were determined by integrating peak areas. The detection limit in xylem sap samples was approximately 20 nM.

Data analysis and speciation calculations

Mean values are presented with ± 1 s.e. Statistical significance of differences between groups was determined using the non-parametric Mann-Whitney *U*-test (also known as the Wilcoxon rank sum test) and differences were reported as significant for values of $P < 0.05$.

GEOCHEM-PC v. 2.0 (Parker et al. 1995) was used to estimate metal free-ion activities and EDTA speciation in the initial nutrient solutions. GEOCHEM-PC is an equilibrium speciation model that uses metal-ligand stability constants from the Version 5.0 NIST 1998 database (NIST 1998).

Results

Uptake of single metal-EDTA complexes

Six-week-old *B. juncea* plants were exposed to nutrient solutions containing 1000 μM PbEDTA²⁻ (with 100 μM excess EDTA) for 60 h. According to equilibrium predictions, [Pb²⁺] in the solution was 0.96 nM. None of the plants

exhibited visible physiological damage to their leaf tissues following exposure to the experimental solutions. The xylem sap concentrations of PbEDTA^{2-} ranged from 2 to 16 μM with an average of $7.1 \pm 1.8 \mu\text{M}$, or 0.71% of the concentration present in the nutrient solution. The range of values in these data highlights the variability among plants grown simultaneously under the same conditions. The speciation of Pb in the xylem sap samples was dominated by EDTA-bound Pb: $85 \pm 8\%$ of total Pb (Pb_{tot}) in the xylem sap could be accounted for as PbEDTA^{2-} . The concentrations of Pb_{tot} and PbEDTA^{2-} were highly correlated with a nearly 1:1 relationship (Fig. 1). In a second set of plants simultaneously exposed to nutrient solutions containing 50 μM PbEDTA^{2-} , PbEDTA^{2-} was detected in most xylem sap samples but could not be reliably quantified. Xylem sap total Pb concentrations ranged from 0.2–1.5 μM , with an average of $0.83 \pm 0.20 \mu\text{M}$, while PbEDTA^{2-} concentrations generally appeared to contribute 20–40% of the total Pb, suggesting that at lower exposure levels, the contribution of PbEDTA^{2-} to overall Pb uptake was less than at higher exposure levels. However, accurately quantifying this difference may require a more sensitive analytical method.

The concentration of Pb_{tot} in leaf tissue increased linearly as a function of xylem sap Pb_{tot} in plants exposed to 50 and 1000 μM PbEDTA^{2-} (Fig. 2). The range of Pb concentrations in leaf tissues was 18–180 $\mu\text{g g}^{-1}$ DW and 88–2500 $\mu\text{g g}^{-1}$ DW for 50 and 1000 μM PbEDTA^{2-} exposure, respectively. The strong correlation between xylem sap Pb and leaf Pb suggests that xylem sap Pb content is a good indicator of shoot Pb accumulation.

Simultaneous uptake of metal–EDTA complexes and PTS

To compare the uptake of three metal–EDTA complexes and the fluorescent tracer PTS, *B. juncea* plants were exposed for 48 h to solutions containing 400 μM each of PbEDTA^{2-} , CdEDTA^{2-} and FeEDTA^{2-} and 380 μM PTS. Exposing each plant to several solutes simultaneously permitted direct comparison of their relative uptake within the same plant while minimizing the effects of biological variability or toxicity of the different metal complexes. Leaf damage was rated as severe ($n = 2$), moderate ($n = 5$) or minimal ($n = 6$) based on the extent of necrotic lesions. Twelve plants produced sufficient

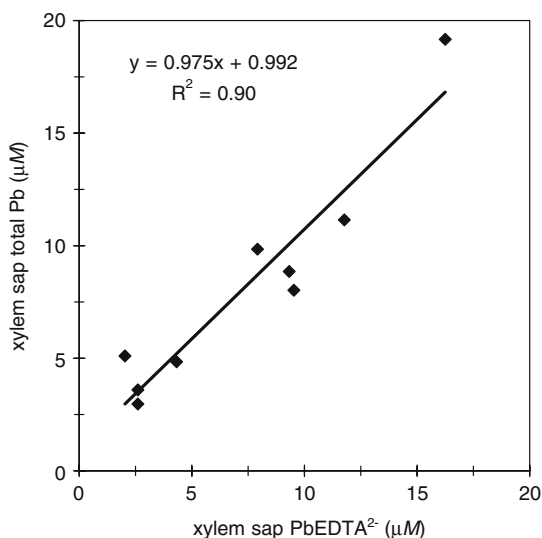


Fig. 1 Total Pb and PbEDTA^{2-} concentrations in the xylem sap of *B. juncea* plants exposed to 1000 μM PbEDTA^{2-} solution for 60 h

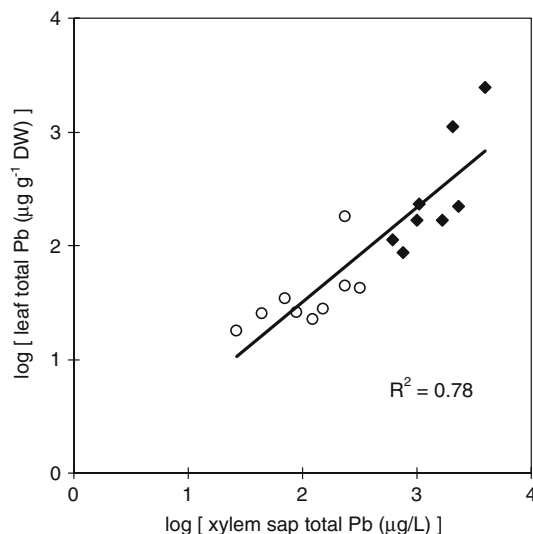


Fig. 2 Total Pb concentrations in the xylem sap ($\mu\text{g/L}$) and leaf tissues ($\mu\text{g g}^{-1}$ DW) of plants exposed to 50 μM (○) and 1000 μM (◆) PbEDTA^{2-} for 60 h

xylem sap for analysis; one “severe” plant produced no xylem sap.

Ten of the twelve plants exhibited similar xylem sap concentrations for each metal species, with one “moderate” plant and one “severe” plant exhibiting concentrations up to 10 times greater than the other plants. Differences between mean concentrations in the “minimal” and “moderate” groups were not statistically significant. The sum of measured metal–EDTA concentrations was $93 \pm 16\%$ of measured total EDTA concentrations, indicating good agreement between the various measurement techniques. The total metal concentrations typically equaled or exceeded their respective metal–EDTA concentrations. Uptake of EDTA-complexed metal contributed more to overall uptake of Pb and Cd than to overall Fe uptake. FeEDTA^- accounted for only $65 \pm 7\%$ of total Fe, while PbEDTA^{2-} accounted for $99 \pm 7\%$ of total Pb and CdEDTA^{2-} accounted for $89 \pm 12\%$ of total Cd. Comparing uptake of the three metal–EDTA complexes shows that xylem sap FeEDTA^- and PbEDTA^{2-} concentrations were linearly related with nearly a 1:1 relationship (Fig. 3). In contrast, CdEDTA^{2-} was present at higher concentrations in the xylem sap than FeEDTA^- and PbEDTA^{2-} in 10 of the 12 samples. The average

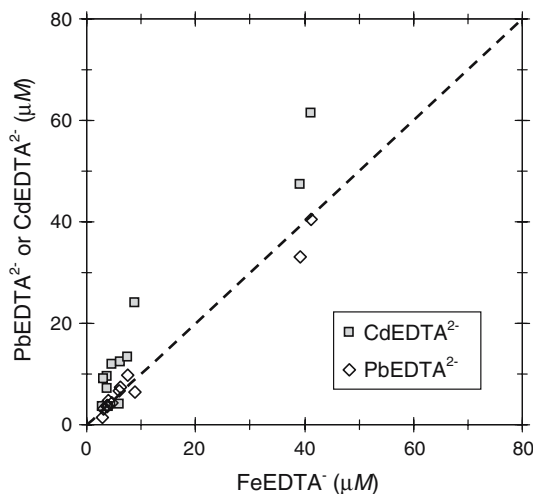


Fig. 3 PbEDTA^{2-} ($\diamond$) and CdEDTA^{2-} ($\square$) concentrations as functions of FeEDTA^- concentration in the xylem sap of *B. juncea* exposed to nutrient solutions containing $400 \mu\text{M}$ each of PbEDTA^{2-} , CdEDTA^{2-} and FeEDTA^- for 48 h. The dotted line represents a 1:1 correlation

$\text{PbEDTA}^{2-}/\text{FeEDTA}^-$ ratio was 0.98 ± 0.07 , while the average $\text{CdEDTA}^{2-}/\text{FeEDTA}^-$ ratio was 1.8 ± 0.2 .

Uptake of PTS, as measured by xylem sap concentrations, was consistently lower than uptake of metal–EDTA (Fig. 4). After normalizing the solute concentration in the xylem sap (C_x) relative to the concentration in solution (C_s), the results suggest a biphasic relationship. At low xylem sap concentrations ($\text{PTS } C_x/C_s < 0.3\%$), PTS and metal–EDTA concentrations were linearly related, with relative concentrations of metal–EDTA 4–10 times greater than those of PTS (inset, Fig. 4). In the two samples with much higher xylem sap concentrations ($\text{PTS } C_x/C_s > 4\%$), relative PTS concentrations were much closer to those of the metal–EDTA complexes, ranging from 40 to 75% of the metal–EDTA concentrations, suggesting less selective uptake of solutes by these plants.

Leaf Pb concentrations were compared to predicted concentrations based solely on xylem sap Pb concentration, total leaf biomass and bulk transpiration (as measured by water loss from solution) over the two-day exposure period. Predicted leaf Pb concentrations ranged from $18\text{--}480 \mu\text{g g}^{-1} \text{DW}$, compared to the range of actual leaf concentrations of $39\text{--}800 \mu\text{g g}^{-1} \text{DW}$. On average, the ratio of predicted:measured leaf Pb concentrations was $80 \pm 38\%$ (range: $47\text{--}180\%$). This comparison shows good agreement between actual and predicted concentrations given implicit assumptions such as constant xylem sap concentrations throughout the entire exposure period.

Comparison of solution- and sand-cultured plants

To investigate the influence of mechanical impedance on root morphology and solute uptake, a side-by-side experiment was performed with plants grown under solution- and sand-culture conditions. The roots of solution-grown *B. juncea* plants were finer, whiter and straighter than those of plants grown under sand conditions. Following a two-day exposure to experimental solutions containing up to $1000 \mu\text{M}$ metal–EDTA, no plants exhibited any visible damage to their leaves. Mean shoot biomass for solution-grown

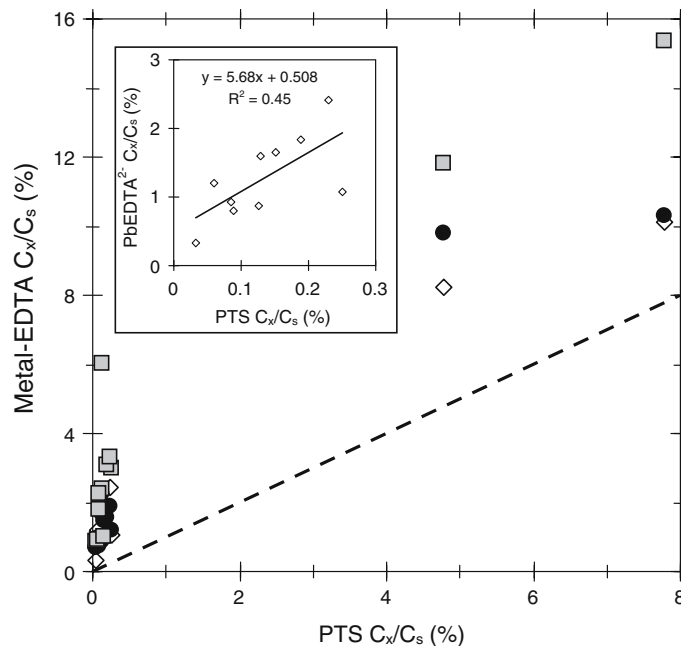


Fig. 4 PbEDTA²⁻ (◇), FeEDTA⁻ (●) and CdEDTA²⁻ (□) concentrations as functions of PTS concentration in the xylem sap of *B. juncea* exposed to nutrient solutions containing 400 μM each of PbEDTA²⁻, FeEDTA⁻ and CdEDTA²⁻ and 380 μM PTS. Concentrations are

expressed as the ratio of xylem sap concentration (C_x) to solution concentration (C_s) in %. Plants were exposed to experimental solutions for 48 h. The dotted line represents a 1:1 correlation. Inset shows C_x/C_s values for PbEDTA²⁻ and PTS for samples in the low range of values

plants was 6.5 ± 0.4 g, which was significantly higher ($P < 0.001$) than the mean shoot biomass for the sand-grown plants, 4.8 ± 0.2 g. This difference may be partly attributable to slightly longer growth time for the plants grown in solution.

In plants simultaneously exposed to 500 μM of both PbEDTA²⁻ and FeEDTA⁻, sand- and solution-grown plants had similar xylem sap composition (Fig. 5). There were no significant differences in the xylem sap concentrations of total Pb, total Fe, PbEDTA²⁻, FeEDTA⁻ or total EDTA between the solution- and sand-grown plants. Consistent with the results described above, PbEDTA²⁻ contributed more to total Pb uptake than FeEDTA⁻ contributed to total Fe uptake ($P < 0.001$, data for sand and solution plants pooled). $110 \pm 9\%$ of total Pb was present as PbEDTA²⁻, whereas only $52 \pm 9\%$ of total Fe was present as FeEDTA⁻. Xylem sap concentrations of FeEDTA⁻ and PbEDTA²⁻ were not as strongly correlated as they were in the previous

experiment ($r^2 = 0.58$) but were not significantly different (data for sand and solution plants pooled). Mean PbEDTA²⁻ and FeEDTA⁻ concentrations were 0.28 and 0.36% of their concentrations in solution, respectively.

Among plants exposed to 1000 μM PbEDTA²⁻, the xylem sap of solution-grown plants contained significantly higher concentrations of PbEDTA²⁻ and total Pb than the xylem sap of sand-grown plants ($P < 0.05$, Fig. 6). Mean PbEDTA²⁻ concentrations for sand- and solution-grown plants were 3.2 ± 1.4 μM and 6.9 ± 1.1 μM, respectively. Only a small portion of this factor of 2 difference in PbEDTA²⁻ concentration may be attributable to changes in the nutrient solutions over the course of the experiment. By the end of the two-day exposure, the PbEDTA²⁻ concentrations in the solution-only buckets had increased by 20% due to transpiration, while concentrations in the sand solutions remained constant, likely a result of the larger reservoir size (120 l for sand culture compared to 8 l for the solution culture).

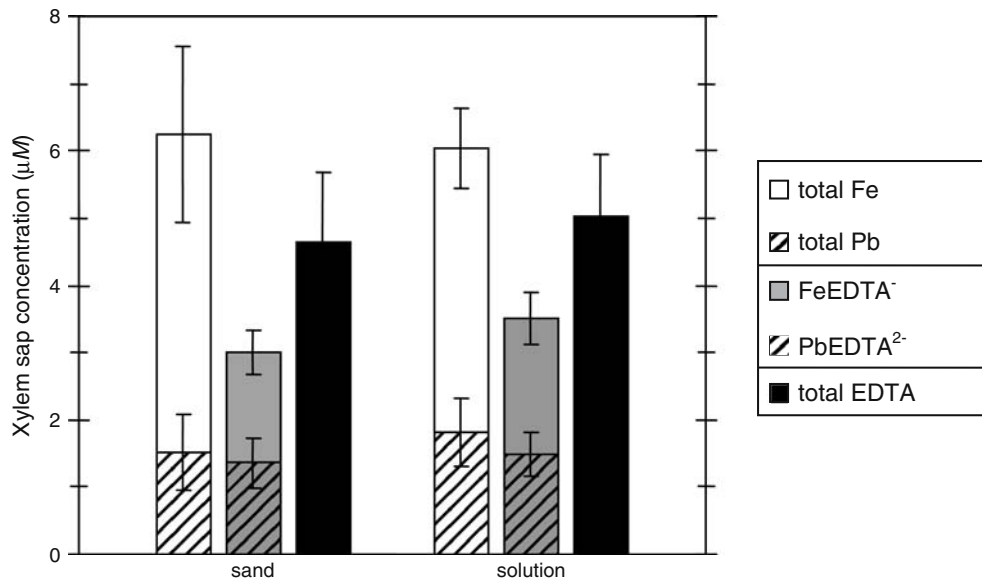


Fig. 5 Speciation of EDTA, Fe and Pb in the xylem sap of *B. juncea* grown in sand ($n = 8$) and solution ($n = 6$) conditions exposed for 48 h to nutrient solutions containing 500 μM FeEDTA^- and 500 μM PbEDTA^{2-} . Error bars represent ± 1 s.e

However, this increase in solution PbEDTA^{2-} concentration alone does not account for the fact that PbEDTA^{2-} concentrations in the xylem sap of solution-grown plants were on average more than double those of the plants grown in sand, suggesting that there may have been different physiological responses depending on growth conditions.

Shoot Pb concentrations increased as a function of solution PbEDTA^{2-} concentration, while shoot

Fe concentrations generally remained constant. In addition to the plants exposed to 1000 μM PbEDTA^{2-} and 500 μM each of FeEDTA^- and PbEDTA^{2-} , another set of plants was simultaneously exposed to solutions containing either 100 μM PbEDTA^{2-} or 50 μM each FeEDTA^- and PbEDTA^{2-} (xylem sap metal–EDTA concentrations were generally below the detection limit and therefore were not reported above). Shoot Pb varied nearly linearly with the total Pb concentrations in solution (Fig. 7), with the exception of a sharp increase in solution-grown plants at 1000 μM PbEDTA^{2-} , resulting in a concentration of $79 \pm 20 \mu\text{g g}^{-1}$ DW. Total shoot Fe concentrations did not increase as solution Fe concentrations increased from 50 μM to 500 μM (data not shown), with most plant shoot Fe concentrations falling in the range of 40–90 $\mu\text{g g}^{-1}$ DW. The range of shoot Fe concentrations observed is consistent with measurements of shoot Fe concentrations in control plants (exposed to 10 μM FeEDTA^-) from previous experiments (data not shown) and is within the range of normal Fe concentrations. It should be noted that all plants were exposed to 10 μM FeEDTA^- during their entire cultivation prior to exposure to experimental solutions, while

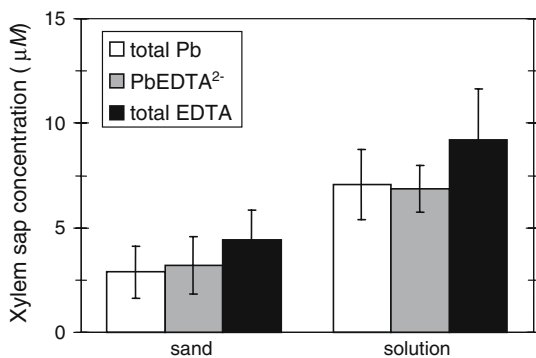


Fig. 6 Speciation of EDTA and Pb in the xylem sap of *B. juncea* grown in sand and solution conditions exposed for 48 h to nutrient solution containing 1000 μM PbEDTA^{2-} ($n = 8$). Error bars represent ± 1 s.e

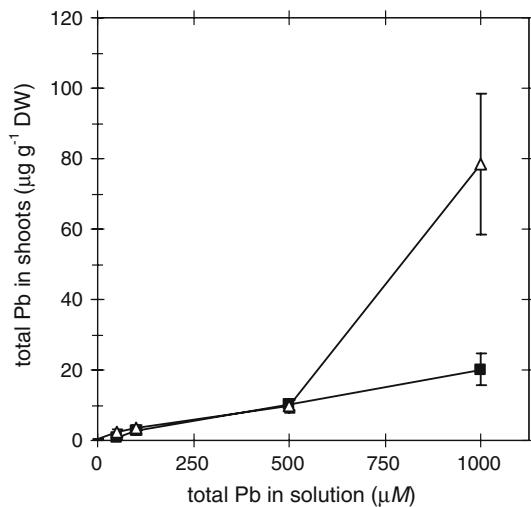


Fig. 7 Total Pb in shoot tissue from sand-grown (closed symbols) and solution-grown (open symbols) plants exposed to solutions containing 50 µM FeEDTA⁻ and 50 µM PbEDTA²⁻, 100 µM PbEDTA²⁻, 500 µM FeEDTA⁻ and 500 µM PbEDTA²⁻, or 1000 µM PbEDTA²⁻. Error bars represent ± 1 s.e. ($n = 5-8$)

plants were only exposed to Pb during the experimental period.

Physiological effects

To measure physiological changes induced by exposure to metal-EDTA solutions, stomatal

conductance and leaf transpiration rates were measured in a subset of control plants and plants exposed to 500 µM FeEDTA⁻ and 500 µM PbEDTA²⁻ under both solution and sand conditions (Table 1). Solution-grown plants were more adversely affected by exposure to the metal-EDTA solutions than sand-grown plants. Exposure to metal-EDTA solutions resulted in significantly lower stomatal conductance relative to controls in solution-grown plants ($P < 0.01$), but not in sand-grown plants. Among plants exposed to the high metal-EDTA solutions, rates of leaf transpiration were significantly lower in solution-grown plants compared to sand-grown plants ($P < 0.05$). These results indicate that although there were no visible changes to the plants following exposure to metal-EDTA solutions, physiological changes did occur.

To determine whether exposure to high metal-EDTA concentrations affected the contribution of apoplastic flow under solution-culture conditions, a separate set of 18 *B. juncea* were exposed to nutrient solutions containing 380 µM PTS and either 0 (control) or 1000 µM PbEDTA²⁻ (Table 2). The average xylem sap PTS concentration was not significantly different between the two groups and there were no significant differences in either the average volume of water transpired or the leaf water content. None of the plants showed visible leaf damage following

Table 1 Stomatal conductance and transpiration rates for solution- and sand-grown *B. juncea* exposed to nutrient solutions containing 500 µM PbEDTA²⁻ and 500 µM FeEDTA⁻ (“high metal-EDTA”) or solutions with 10 µM FeEDTA⁻ (“control”)

	Sand	Solution
<i>Stomatal conductance (mmol m⁻² s⁻¹)</i>		
Control	460 $\pm$ 9 ^{ab}	466 $\pm$ 16 ^a
High metal-EDTA	424 $\pm$ 11 ^{ab}	396 $\pm$ 16 ^b
<i>Transpiration (mmol m⁻² s⁻¹)</i>		
Control	6.92 $\pm$ 0.10 ^{cd}	6.75 $\pm$ 0.31 ^{cd}
High metal-EDTA	7.03 $\pm$ 0.09 ^c	6.56 $\pm$ 0.19 ^d

Data presented as mean $\pm$ s.e. ($n = 5-6$). Significance determined using Mann-Whitney *U*-test, and data that are significantly different ($P < 0.05$) do not share superscripted letters

Table 2 Xylem sap concentrations of PTS and total EDTA, bulk transpiration and leaf water content for *B. juncea* plants exposed to either 0 or 1000 µM PbEDTA²⁻ for 72 h. The concentration of PTS in solution was 380 µM

	0 µM PbEDTA ²⁻	1000 µM PbEDTA ²⁻
<i>n</i>	9	8
Xylem sap PTS (µM)	0.19 $\pm$ 0.06	0.10 $\pm$ 0.05
Xylem sap total EDTA (µM)	n.a.	3.4 $\pm$ 0.7
Bulk transpiration (L H ₂ O g ⁻¹ dry leaf mass)	0.13 $\pm$ 0.01	0.12 $\pm$ 0.01
Leaf water content (% m/m)	92.0 $\pm$ 0.2	92.2 $\pm$ 0.3

Results presented as mean $\pm$ 1 s.e. Total EDTA concentrations were not analyzed for control plants. No significant differences were observed between the two treatments

exposure. In the plants exposed to 1000 μM PbEDTA^{2-} , the C_x/C_s values for PTS were 0.027 ± 0.014 %, while C_x/C_s values for total EDTA were 0.34 ± 0.07 %, consistent with our earlier findings of greater relative uptake of EDTA than PTS.

Discussion

The results of this study suggest that there are two stages of uptake of hydrophilic compounds by plants. Under normal conditions, the xylem sap concentrations are relatively low (i.e., PTS $C_x/C_s < 0.5\%$) and there is a high degree of selectivity among different solutes. At these low concentrations, xylem sap metal–EDTA concentrations, relative to their concentration in solution, were 4–11 times higher than those of PTS (Fig. 4). Under conditions of stress, however, xylem sap concentrations are significantly higher (i.e., PTS $C_x/C_s > 3\%$) and the selectivity among solutes decreases. At these higher concentrations, relative concentrations of PTS and metal–EDTA in xylem sap samples agreed to within a factor of two.

In the absence of stress, a high degree of selectivity in uptake also is apparent in differences in the uptake of various metal–EDTA complexes. Bell et al. (2003) compared uptake of ^{14}C -labelled EDTA from solutions that contained EDTA primarily as CaEDTA^{2-} , FeEDTA^- or ThEDTA^0 and observed that EDTA uptake into leaves decreased with increasingly negative charge of the metal–EDTA complex. The authors attributed these variations to differences in interactions with cell wall membranes and the ability of the complexes to be transported by endodermal passage cells. Based on these findings, one might predict that PbEDTA^{2-} and CdEDTA^{2-} would exhibit similar patterns of uptake and that FeEDTA^- should be present at higher concentrations. However, the results of our study demonstrate that PbEDTA^{2-} and FeEDTA^- concentrations were nearly identical in the xylem sap of plants exposed to equimolar Pb- and Fe–EDTA solutions, while CdEDTA^{2-} was consistently present at slightly higher concentra-

tions. The similarity in FeEDTA^- and PbEDTA^{2-} uptake was observed despite the fact that total Fe in xylem sap was greater than total Pb. Because Fe is a micronutrient, it is reasonable that membrane-mediated uptake of other forms of Fe would be greater than uptake of a non-nutrient metal such as Pb. It is also possible that some of the Fe present in the xylem sap was taken up prior to the beginning of the metal–EDTA exposure.

These patterns of uptake suggest that subtle differences in the structure of the metal–ligand complex may be more important than the overall charge of the complex. The differences may be dependent on a combination of size, charge, stability and the degree of complexation with hexadentate EDTA. In the current study, monovalent and divalent EDTA complexes did exhibit greater uptake than trivalent PTS; however, Skinner and Radin (1994) observed that xylem sap concentrations of PTS were approximately 10 times lower than those of the closely related PTS_4 , which carries a (–4) charge, again supporting the idea that there is not a simple relationship between charge and uptake of similar compounds. Although the magnitude of PTS and metal–EDTA uptake was different, their uptake was linearly related for values of $C_x/C_s < 0.5\%$, suggesting that the routes of uptake are the same, even though their relative mobility may be different.

The observed increases in xylem sap PTS and metal–EDTA concentrations at higher levels of stress, also evident as moderate and severe necrosis in leaf tissues, are consistent with previous studies demonstrating increased bypass flow under conditions of chemical and physical stress. Typically, PTS C_x/C_s values for healthy plants range from 0.01 to 0.3 % (Hanson et al. 1985; Moon et al. 1986; Skinner and Radin 1994; Yeo et al. 1987; Zimmermann and Steudle 1998). By comparison, anaerobic conditions (Hanson et al. 1985) and transplanting (Moon et al. 1986; Yeo et al. 1987) have led to significant increases in PTS C_x/C_s values. Transplanting also led to enhanced metal accumulation from contaminated soil treated with EDTA (Wu et al. 1999).

In our study, metal–EDTA complexes were associated with phytotoxicity in some plants,

although the exact mechanism of this damage is not known. Excess chelating ligand previously has been invoked as a possible explanation. Vassil et al. (1998) observed significantly enhanced Pb uptake from a PbEDTA^{2-} solution when total EDTA exceeded Pb by at least 0.25 mM, suggesting that free and/or weakly-complexed EDTA destabilized root cell membranes and allowed unrestricted movement of solutes into the cytosol. Collins et al. (2002) concluded that *B. juncea* is particularly sensitive to the effects of excess EDTA compared to barley and potato plants. The authors also observed a biphasic pattern of ZnEDTA^{2-} uptake, with relatively low $[\text{ZnEDTA}^{2-}]$ in the xylem at low levels of EDTA addition to the soil, presumably because roots were intact and undamaged, and disproportionately high xylem sap $[\text{ZnEDTA}^{2-}]$ resulting from exposure to high concentrations of EDTA, implying significant root damage. However, in our study, EDTA concentrations exceeded transition metal concentrations by only 50–100 μM , resulting in a concentration of weakly complexed EDTA (primarily CaEDTA^{2-}) that is unlikely to induce significant physiological stress. Direct damage by intact metal–EDTA complexes is also possible, although little is known in general about the effects of chemical stress on the endodermis. Chlorate has been found to cause damage to the endodermis of rice seedlings, although the nature of this damage is not well understood (Borges et al. 2004). However, exposure to 1000 μM metal–EDTA did not necessarily lead to increased uptake of PTS (Table 2), suggesting no consistent effect of metal–EDTA exposure on the contribution of bypass apoplastic flow. From our results, it is not possible to determine whether increased xylem sap concentrations and decreased selectivity are attributable to a greater proportion of apoplastic bypass flow, perhaps through breakdown of the endodermal barrier, or if root cell membrane integrity is compromised by exposure. While EDTA alone is hypothesized to affect root cell membranes by depleting membrane-bound Ca (Foote and Hanson 1964), there is little direct evidence that metal–EDTA complexes operate by a similar mechanism.

Effects of growth conditions

Root morphology is commonly recognized to differ between hydroponic and soil-grown plants (Parker and Norvell 1999), but little is known about how these differences affect solute uptake. Using sand to simulate the mechanic impedance associated with soil, our results demonstrate that metal–EDTA uptake by solution-grown plants is equal to (Fig. 5) or greater than (Fig. 6) uptake in plants grown in sand. The physiological explanation for greater PbEDTA^{2-} uptake at the 1000 μM level by plants grown in solution may involve differences in water availability. Decreased water availability, caused by culturing aeroponically or in vermiculite, has been associated with enhanced suberization of the endodermis as well as the formation of an exodermis (Schreiber et al. 1999; Zimmermann et al. 2000). These changes could lead to restricted solute mobility via the apoplastic bypass flow pathway. Although the sand was saturated five times per day with nutrient solution, the roots may have experienced more water stress than the plants grown in solution. If the sand-grown plants did develop thicker endodermal barriers, then they would have had better physical barriers against metal–EDTA uptake. However, the xylem sap concentrations in plants at the 500 μM combination exposure were nearly identical under both sets of conditions, suggesting that morphological differences did not significantly affect solute uptake and apoplastic flow.

Alternatively, the sand-grown plants may have been more resistant to physiological stress than the solution-grown plants. The significant increase in Pb accumulation from the 500 to 1000 μM PbEDTA^{2-} exposure in solution-grown plants may have been caused by PbEDTA^{2-} induced root damage. Consistent with the hypothesis that sand-grown plants are less susceptible to PbEDTA^{2-} phytotoxicity, previous studies have demonstrated that solution-grown plants are equally or more sensitive to chemical stresses such as NaCl and Al than their sand-grown counterparts (Horst et al. 1990; Villagarcia et al. 2001; Zaiter and Mahfouz 1993). In plants

exposed to 500 μM of both PbEDTA^{2-} and FeEDTA^- , only solution-grown plants exhibited significant decreases in stomatal conductance.

Overall, the sand versus solution culture experiments demonstrate that while solution-grown plants at times exhibit greater metal–EDTA uptake, sand and solution-cultured plants generally exhibit similar uptake of metal–EDTA complexes. Consequently, it is reasonable to predict that at similar soil solution concentrations of metal–EDTA complexes, soil-grown plants would exhibit similar patterns of metal–EDTA uptake.

Implications

While the free-ion activity model (FIAM) can be useful for predicting metal uptake, there are a number of situations in which metal–ligand complexes likely will play a significant role in metal uptake by plants. Chelator-buffered nutrient solutions were designed to mimic soil solutions, in which a relatively large pool of complexed metal buffers constant metal free-ion activity (Parker and Norvell 1999). These solutions have widely been used to study plant nutrition and toxicity, but they rely on the premise that chelated metals are biologically unavailable. While free metal ions may form the majority of biologically active metal taken up by membrane-mediated active transport, metal–ligand complexes may play an important role in overall metal uptake and accumulation, particularly for non-nutrient metals such as Pb and Cd. Direct uptake and accumulation of ZnDTPA was proposed to explain tomato plants that were markedly Zn-deficient, yet had physiologically adequate concentrations of total Zn in their shoots (Parker et al. 1992). While previous examples of exceptions to the FIAM in the presence of chelating agents may be caused by other factors, such as overcoming diffusive limitations of free metal ions at the root surface (e.g., Degryse et al. 2006), the direct uptake of metal–ligand complexes also should be considered in the interpretation of metal uptake from chelator-buffered solutions.

The importance of chelated metals is even more broadly relevant in chelate-enhanced phytoextraction, where metal-binding ligands, such as EDTA, promote desorption from mineral

surfaces and organic matter and maintain a large pool of metal–ligand complexes in the soil solution. The range of concentrations used in our study is consistent with those measured in the soil of phytoextraction experiments. For instance, Collins et al. (2001) measured ZnEDTA^{2-} concentrations up to 830 μM in soil solutions following addition of 2.5 mmol EDTA kg^{-1} soil, an application rate similar to those used in phyto-remediation studies (Blaylock et al. 1997; Wu et al. 1999). Since metal–EDTA complexes can dominate soil solution metal speciation, the results of our study imply that it is likely that chelated metals will comprise the majority of metals taken up by plants during chelate-enhanced phytoextraction. Our results suggest that physiological stress could increase the unrestricted uptake of metal–chelate complexes, provided that the stress does not significantly decrease transpiration rates. With an improved understanding of the relationships between metal speciation and uptake, it may be possible to develop safer and more effective chelate-enhanced phytoextraction systems.

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